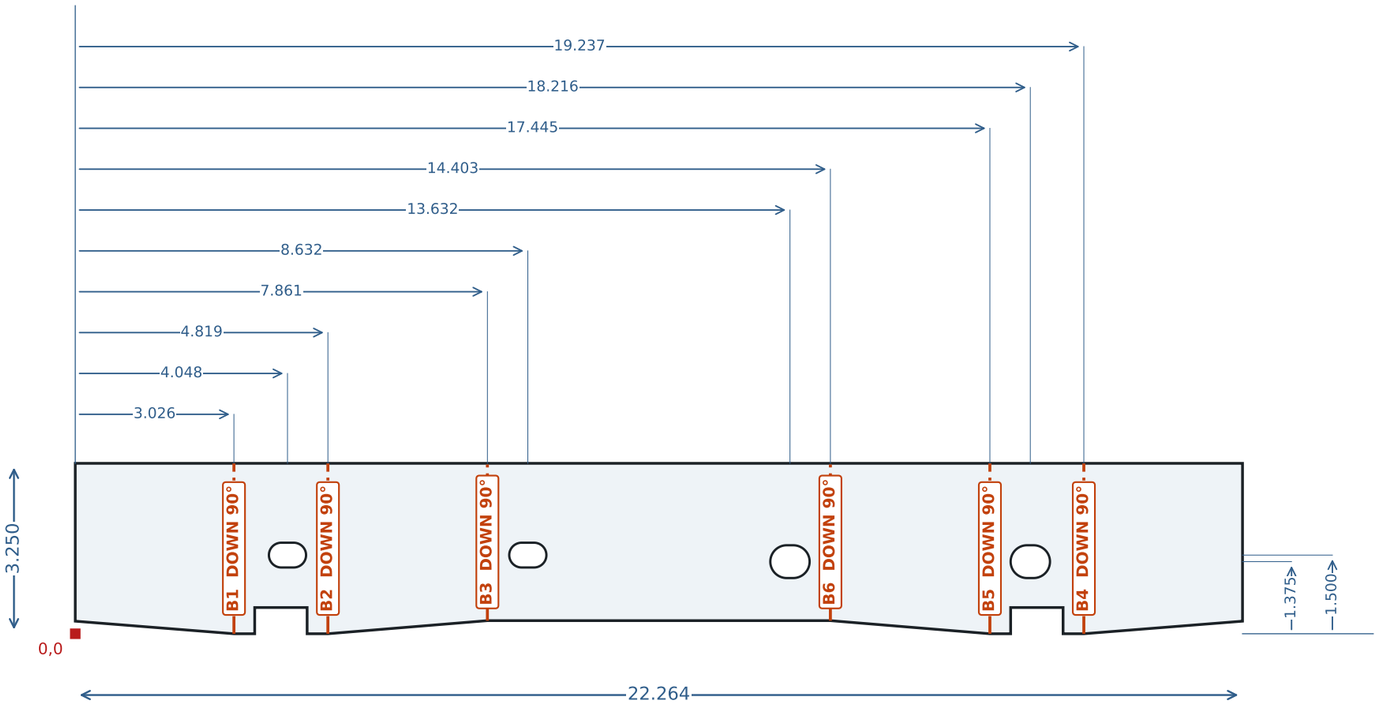
FR-2 · FR-2 one piece (top + 2 U legs)

11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS · flat blank 22.264″ × 3.250″ · ≈ 2.27 lb each · QTY 1 FR + 1 FL = 2 per vehicle

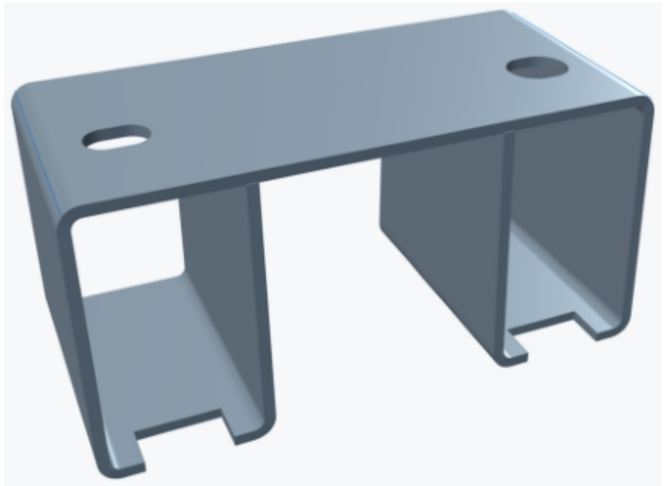
MATERIAL: HR steel sheet A1011 CS
THICKNESS: 11 ga t = 0.1196″ (3.04 mm)

FLAT PATTERN (as laser-cut, marked side up)



FR-2 · 11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS

FORMED PART (after bending)



(shape; thickness now 11 ga 0.120″)

BEND TABLE

Bend	Angle	Direction	Inside R	Order	Bend line position
B1	90.0°	DOWN (away)	R=t 0.120	1	X = 3.026
B2	90.0°	DOWN (away)	R=t 0.120	3	X = 4.819
B3	90.0°	DOWN (away)	R=t 0.120	5	X = 7.861
B4	90.0°	DOWN (away)	R=t 0.120	2	X = 19.237
B5	90.0°	DOWN (away)	R=t 0.120	4	X = 17.445
B6	90.0°	DOWN (away)	R=t 0.120	6	X = 14.403

HOLES (centre from 0,0)

Hole	Size	X	Y	Made by
H1	cut-out 0.708×0.472	4.048	1.500	laser
H2	cut-out 0.709×0.472	8.632	1.500	laser
H3	cut-out 0.750×0.625	13.632	1.375	laser
H4	cut-out 0.750×0.625	18.216	1.375	laser

FILES

- FR-2_FR_cut.dxf (right, qty 1 / vehicle)
- FR-2_FL_cut.dxf (left = mirror, qty 1 / vehicle)
- FR-2_ref.dxf (reference: + bend lines, drill-later, material notes)

NOTES

1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
2. MATERIAL HR steel sheet A1011 CS, 11 ga t = 0.1196″ (3.04 mm). Bend allowance (K = 0.44, inside radius = t) is already in the flat.
   Bend on the orange lines; UP = toward you when the part lies as drawn.
3. Holes marked DRILL LATER (red dashed) are not laser cut - drill after bending/welding.
4. Deburr all edges, remove dross.
5. Right part (FR) shown. Left part (FL) = mirror: cut *_FL_cut.dxf, bend the same UP/DOWN.

* one blank, 6 bends 90° the same way (R = t): inner walls first, then the bottoms, then the outer walls; all ovals (top and leg bottoms) laser cut;
½" notch at the front edge of each bottom; inner walls reach the underside of the top (11 ga)

FR-2 Bridge bracket (top + 2 U legs)	**HR steel sheet A1011 CS 11 ga 0.120″ (3.04 mm)**	Units: inch · NOT TO SCALE	Sienna WAV frame	**REV B**
FR-2 · FR-2 one piece (top + 2 U legs)	**QTY 1 FR + 1 FL = 2 per vehicle**	Tol.: cut ±0.010″ · bend ±1° · IR = t	2026-10-31	**SHEET 1/1**